Eman Tehseen

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Computer Engineering student at COMSATS University with hands-on experience in embedded systems, digital design, machine learning, and full-stack web development. Worked on FPGA, AI, and software development projects through academic coursework and internships.

ACADEMIC QUALIFICATION

COMSATS University Islamabad, Abbottabad Campus — Abbottabad,Pakistan
Bachelor of Science in Computer Engineering — *Sep 2023 – On-going*

EXPERIENCE

National Radio & Telecommunication Corporation (NRTC) — Haripur, Pakistan
Intern, Intelligence Division Lab — *Aug 2025*

- Developed a Weather Application using React.js and integrated external APIs for real-time weather information retrieval.
- Built a Rock Paper Scissors game in C++ featuring GUI components and score-tracking mechanisms.
- Designed and implemented a User Authentication API using FastAPI, MongoDB, JWT authentication, and bcrypt password hashing.
- Developed a Social Media Desktop Application using Python Tkinter with user authentication and posting functionality.
- Designed and simulated a 16-bit Full Adder using Multisim and embedded system measurement tools using Arduino and Proteus.

Technik Nest Pvt. Ltd.
Web Development Intern / Trainee

- Completed professional training in modern web development technologies and full-stack application development.
- Developed and deployed CampusConnect, a university management and collaboration platform using Next.js, React.js, Supabase, PostgreSQL, and Vercel.
- Worked with responsive UI design, authentication systems, database integration, cloud deployment, and version control using GitHub.

ACADEMIC & PERSONAL PROJECTS

Brain Tumor MRI Classification Using Transfer Learning & Deep Learning | *TensorFlow, CNN, ResNet50, EfficientNetB3, Python*

- Developed an automated MRI classification system capable of identifying Glioma, Meningioma, Pituitary Tumor, and No Tumor classes.
- Implemented transfer learning using ResNet50 and EfficientNet architectures alongside fine-tuning, dropout regularization, batch normalization, and TensorFlow pipelines.
- Built optimized preprocessing pipelines including duplicate image detection, EfficientNet-specific preprocessing, and data pipeline optimization.
- Achieved **98.7% testing accuracy** and **99.4% training accuracy** using an optimized EfficientNetB3 model.

32-bit Single-Cycle RISC-V Processor| *Verilog HDL, ModelSim, Xilinx, Spartan-3 FPGA*

- Designed and implemented a complete RV32I-compatible RISC-V processor architecture in Verilog HDL.
- Developed processor modules including Program Counter, Instruction Memory, Register File, ALU, Data Memory, Control Unit, ALU Control Unit, and PC Logic.

- Simulated and verified processor functionality using ModelSim and synthesized the design using Xilinx tools.
- Successfully deployed and tested the processor on a Spartan-3 FPGA development board.

CampusConnect: University Management & Collaboration Platform | *Next.js, React.js, TypeScript, Supabase, PostgreSQL, Vercel*

- Developed a full-stack university collaboration platform supporting authentication, event management, announcements, team collaboration, and project management.
- Implemented secure authentication and authorization using Supabase Authentication and Row-Level Security (RLS).
- Designed and managed PostgreSQL database schemas for events, projects, teams, announcements, and user profiles.
- Deployed the application on Vercel and maintained source control using GitHub

AI-Based Smart Surveillance System | *Raspberry Pi, OpenCV, Machine Learning*

- Designed and implemented a smart surveillance platform for real-time security monitoring using video analytics.
- Utilized OpenCV and machine learning techniques for fire/smoke detection and unauthorized access classification.
- Deployed the system on Raspberry Pi for edge-based intelligent monitoring.

Text-to-Speech System with DSP Effects | *Python, Pyttsx3, Pydub*

- Developed a speech synthesis application with customizable pitch, speed, and audio effects.
- Integrated digital signal processing techniques for enhanced audio output and usability.

TECHNICAL SKILLS

Programming Languages: C++, Python, Verilog HDL, TypeScript, JavaScript, SQL
Machine Learning & AI: TensorFlow, CNNs, Transfer Learning, EfficientNet, ResNet50, Deep Learning, Computer Vision, OpenCV
Web Development: React.js, Next.js, Tailwind CSS, FastAPI, Supabase, REST APIs
Embedded Systems & Hardware: Arduino, Raspberry Pi, Spartan-3 FPGA, Digital System Design
Tools & Platforms: Git, GitHub, VS Code, Vercel

ADDITIONAL INFORMATION

Languages: English, Urdu
Awards/Activities:

- Named on the **COMSATS Campus Honors List** for maintaining a **4.00 GPA** and outstanding academic performance.
- Successfully completed internship at **National Radio & Telecommunication Corporation (NRTC), Haripur**.
- Successfully completed Web Development Internship / Professional Training at Technik Nest Pvt. Ltd., with hands-on experience in modern full-stack web development technologies

ACTIVITIES & LEADERSHIP:

- Organized sustainability awareness campaigns under the IET and Robotics Club at COMSATS Abbottabad.
- Team Member (Ushers), COMSATS Convocation Organization Committee 2024.
- Active Member, COMSATS Character Building Society; participated in fundraising and awareness initiatives.
- Organizing Team Member, IEEE Student Branch, COMSATS University Abbottabad Campus.